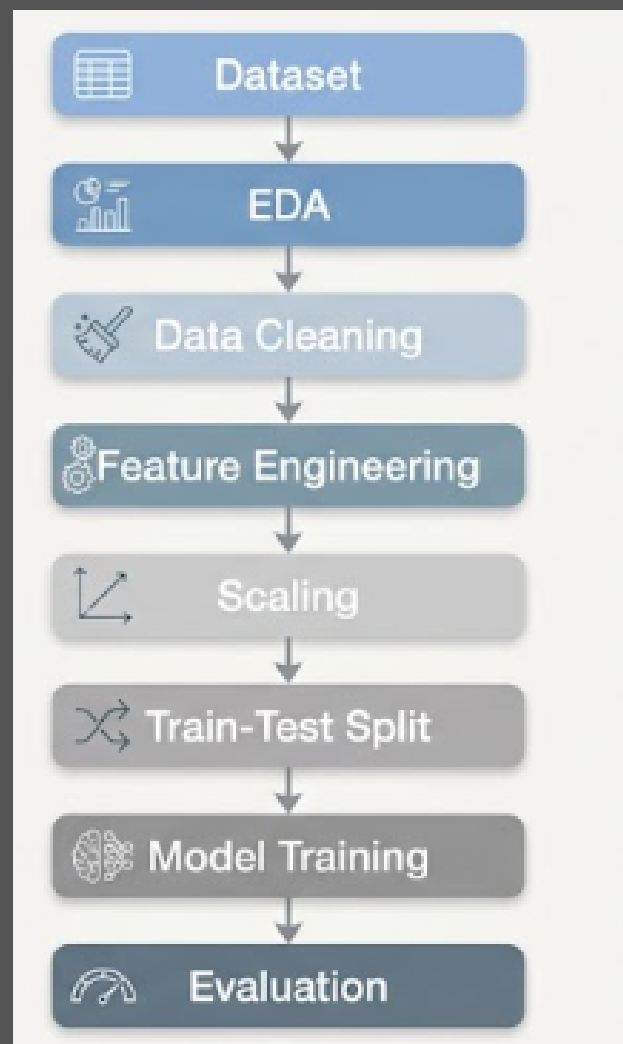


Day 25 - Final Technical Review - Summary

Problem Statement :

In Underwater Optical Communication (UWOC), turbidity (NTU) causes attenuation and scattering of optical signals. This changes the received signal strength, making fixed-threshold detection unreliable. The objective is to use supervised machine learning to classify transmitted bits (0 or 1) accurately under varying turbidity conditions.

Methodology



Models Evaluated:

- Fixed Threshold
- Logistic Regression
- SVM
- Random Forest
- MLP

Limitations :

- Small dataset size
- Moderate class imbalance
- Limited feature set
- Performance degradation at high NTU levels
- No real-world deployment testing

KEY FINDINGS

Dataset



399
SAMPLES

✓ NO MISSING
VALUES

✓ NO DUPLICATE
RECORDS

✓ NO INVALID
BIT LABELS

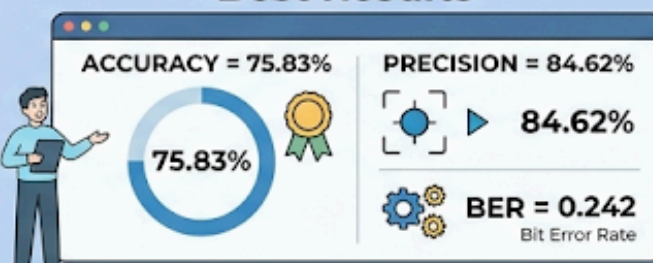


**MODERATE CLASS
IMBALANCE**
(60.65% vs 39.35%)

Best Model



Best Results



Robustness Study



PERFORMANCE REMAINS STRONG

Robustness Study



ACCURACY DECREASES
AS NTU INCREASES

HIGH TURBIDITY
INCREASES SIGNAL
OVERLAP
AND BER

X
BER

NTU → LINK TO TURBIDITY

Future Work

- Collect larger UWOC datasets.
- Test advanced models (XGBoost, LightGBM, Deep Learning).
- Incorporate additional environmental features.
- Evaluate real-time deployment on embedded hardware.
- Improve robustness under high turbidity conditions.

